

G001 Motion System Reference Manual

Motion Platform Architecture

The G001 motion system employs a six-degree-of-freedom (6-DOF) Stewart platform configuration, utilizing electro-hydraulic actuators to provide pitch, roll, yaw, heave, sway, and surge motion cues. The platform architecture consists of three primary subsystems: the hydraulic power unit (HPU), the motion control computer (MCC), and the actuator assembly with integrated position feedback sensors.

Each of the six actuators incorporates a dual-redundant linear variable differential transformer (LVDT) for position sensing, with real-time comparison logic implemented in the MCC to detect discrepancies exceeding $\pm 2\text{mm}$. The servo valves operate at a bandwidth of 50 Hz, providing sufficient response characteristics for transport category aircraft simulation. Actuator stroke length is 1270mm ($\pm 635\text{mm}$ from neutral), with a maximum extension rate of 500mm/sec and a load capacity of 45kN per actuator.

The MCC receives motion drive commands from the host computer via a dedicated 1000BASE-T Ethernet link operating on VLAN 10. Motion cueing algorithms process aircraft state data at 60 Hz, applying washout filters to maintain platform position within operational limits while maximizing perceived motion fidelity. The system supports three operational modes:

1. **FULL MOTION** - All six DOF active with standard washout parameters
2. **REDUCED MOTION** - Limited to $\pm 50\%$ actuator travel for maintenance or training scenarios
3. **FREEZE** - Platform locked at current position, motion commands inhibited
4. **PARK** - Controlled descent to neutral position with all actuators at mid-stroke

Platform status is continuously monitored through the motion system BITE (Built-In Test Equipment), which performs cyclic checks of hydraulic pressure (nominal 3000 PSI ± 100), servo valve response, LVDT continuity, and emergency stop circuit integrity. Any fault condition generates a corresponding code displayed on the instructor operating station (IOS) and triggers appropriate platform safing procedures. Critical faults (hydraulic pressure loss, actuator position error $> 10\text{mm}$, or servo valve failure) initiate an immediate controlled emergency stop sequence.

Motor Driver Boards and Connectors

The G001 motion system employs six independent servo motor driver boards (P/N MD-6400-G) mounted in the main drive cabinet located in the equipment room beneath the simulator platform. Each driver board interfaces directly with a corresponding Stewart platform actuator through shielded multi-conductor cables rated for continuous flexing during motion operation.

The driver boards receive position and velocity commands from the Motion Control Computer (MCC) via the primary CAN bus operating at 1 Mbps. Each board contains an integrated DSP running at 200 MHz for real-time PID loop calculations, with position updates occurring at 2 kHz. Onboard diagnostics continuously monitor motor current, voltage, temperature, and encoder feedback integrity.

Connector Assignments:

- **J1 (DB-37 Male):** Primary power input (270 VDC nominal, 150-300 VDC operating range)
- **J2 (DB-25 Female):** CAN bus interface and discrete I/O signals
- **J3 (MS3106A-22-14S):** Motor phase outputs (U, V, W) and resolver feedback
- **J4 (BNC):** Analog monitor output for oscilloscope diagnostics
- **J5 (RJ-45):** Ethernet service port for firmware updates and configuration

Driver Board Fault Code Verification Procedure:

1. Access the MCC maintenance terminal and navigate to MOTION > DIAGNOSTICS > DRIVER STATUS
2. Select the driver board number (1-6) corresponding to the affected actuator
3. Record any active fault codes displayed in hexadecimal format
4. Cross-reference fault codes with Table 4-2 in Section 4.8
5. Clear latched faults using the RESET FAULTS soft key after correcting the underlying condition
6. Verify fault clearance by commanding a 10% extension test cycle
7. Monitor motor current on J4 using a digital oscilloscope to confirm smooth operation

Each driver board includes redundant temperature sensors with automatic thermal shutdown at 85°C junction temperature. The boards feature conformal coating for humidity protection and operate within ambient temperatures of 0-50°C.

Motion Fault Codes (4xB Series)

The 4xB series motion controllers implement a comprehensive fault detection and isolation system that monitors all six degrees of freedom actuators in real-time. Fault codes are displayed on the Motion Control Unit (MCU) LCD panel and simultaneously logged to the Instructor Operating Station (IOS) event recorder. Each fault code follows the format 4xB-### where 'x' represents the specific actuator axis (S=Surge, W=Sway, H=Heave, R=Roll, P=Pitch, Y=Yaw) and '###' indicates the fault type.

Common 4xB Series Fault Codes:

- **4SB-101:** Surge actuator encoder mismatch exceeding 0.5mm tolerance
- **4WB-102:** Sway actuator hydraulic pressure below 2800 PSI threshold
- **4HB-103:** Heave actuator overtravel detected, upper limit switch engaged
- **4RB-201:** Roll axis servo amplifier thermal shutdown initiated
- **4PB-202:** Pitch actuator resolver fault, position feedback loss
- **4YB-203:** Yaw axis velocity command exceeded 150mm/sec safety limit
- **4MB-301:** Master motion controller CAN bus communication timeout
- **4MB-302:** Emergency stop circuit activated, all axes inhibited
- **4MB-401:** Motion washout filter computation overflow detected
- **4MB-501:** Platform levelness verification failed during BIT sequence

Fault Code Reset Procedure:

1. Verify platform is in neutral position (all actuators at mid-stroke)
2. Access MCU maintenance menu using supervisor password
3. Select "FAULT LOG" from main menu
4. Review active fault codes and timestamps
5. Address underlying mechanical or electrical issue per maintenance manual
6. Select "CLEAR FAULTS" option
7. Confirm fault clearance with IOS operator
8. Execute motion system BIT (Built-In Test) sequence
9. Verify all axes respond correctly to low-amplitude test signals
10. Document fault occurrence and corrective action in aircraft logbook

Critical faults (3xx series) require immediate platform shutdown and cannot be cleared without engineering authorization. Non-critical faults (1xx-2xx series) may be temporarily suppressed for training continuation at instructor discretion, though repeated occurrences mandate maintenance intervention.

Connector Reseating Procedure

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Intermittent motion platform faults, including sporadic actuator position errors, resolver signal dropouts, or transient BIT failures in the motion control system, may be resolved through systematic connector reseating. This procedure addresses mechanical connector degradation resulting from thermal cycling, vibration exposure, or oxidation of contact surfaces within the motion drive electronics (MDE) and actuator interface assemblies.

Prior to initiating connector reseating operations, ensure the motion system is fully powered down and the main circuit breaker panel is locked out and tagged. Verify hydraulic pressure has been relieved to zero PSI on all actuator circuits and that the platform is resting on mechanical safety locks. Discharge any residual capacitance in the servo amplifier banks by waiting a minimum of ten minutes after power removal.

Reseating Sequence

1. Document the current fault codes from the Motion Control Computer (MCC) diagnostic display, noting fault occurrence patterns and affected actuator channels.
2. Beginning with the MDE cabinet, systematically disconnect each D-sub connector at the servo amplifier interface cards, inspecting pins for discoloration, corrosion, or mechanical damage.
3. Apply contact cleaner (MIL-PRF-81309 Type III) to connector pins using lint-free applicators. Allow thirty seconds for solvent evaporation.
4. Firmly reseat each connector, ensuring the locking mechanism fully engages with an audible click. Apply torque specifications per maintenance manual section 2.4.7 for threaded connector shells.
5. Proceed to actuator resolver connectors at each of the six motion axes, repeating the inspection and reseating process.
6. Document connector serial numbers and reseating completion in the maintenance log with technician initials and timestamp.

Following reseating operations, restore power and execute a comprehensive BIT sequence through the MCC diagnostic menu. Monitor resolver feedback signals during a slow-speed motion profile test, verifying position error remains within ± 0.05 degrees across all axes. Persistent faults after reseating indicate component-level failure requiring replacement per troubleshooting decision tree G001-FT-003.

Full Motion Test Checklist

Full Motion Test Checklist

The Full Motion Test Checklist provides a systematic verification sequence for validating all six degrees of freedom (6-DOF) motion platform functionality prior to training operations. This procedure must be completed following any maintenance activity affecting the motion system, after extended shutdown periods exceeding 72 hours, or when motion fidelity discrepancies are reported by instructional staff.

Pre-Test Configuration

Prior to initiating motion tests, verify the following conditions:

- Motion platform hydraulic pressure stabilized at 2,800-3,000 PSI across all actuators
- Accumulator pre-charge pressure within 1,950-2,050 PSI range
- Motion Control Computer (MCC) BIT completed with zero fault codes
- Emergency stop circuits tested and reset
- Simulator weight and balance within certified envelope ($\pm 2\%$ CG tolerance)
- All motion limit switches indicating normal range
- Hydraulic fluid temperature between 100-120°F

Motion Test Sequence

1. **Neutral Position Verification** - Confirm all six actuators report zero extension with platform level within ± 0.1 degrees on all axes.
2. **Individual Actuator Test** - Command each actuator through 25% extension increments while monitoring hydraulic flow rates and servo valve response times. Nominal extension rate: 8-12 inches per second.

SAMPLE / NOT REAL

3. **Axis Isolation Tests** - Execute single-axis commands (pitch, roll, yaw, surge, sway, heave) through full available travel at 50% rate limiting. Record peak accelerations and compare against motion system specification table.
4. **Coordinated Motion Profile** - Run pre-programmed test profile simulating takeoff rotation, moderate turbulence, and landing flare. Verify motion cues align with visual system within 40ms latency threshold.
5. **Emergency Stop Function** - Trigger E-stop during coordinated motion and verify controlled deceleration to neutral position within 2.5 seconds maximum.

Document all test results in the Motion System Maintenance Log, noting any deviations exceeding $\pm 5\%$ of nominal values for engineering review.